

Superstore sales story

Sample Superstore data, 2014 to 2017 (9,994 order lines). Prepared by: Vaishnavi B.Gawande

Project repository: <https://github.com/twinklegawande169-sketch/superstore-capstone>

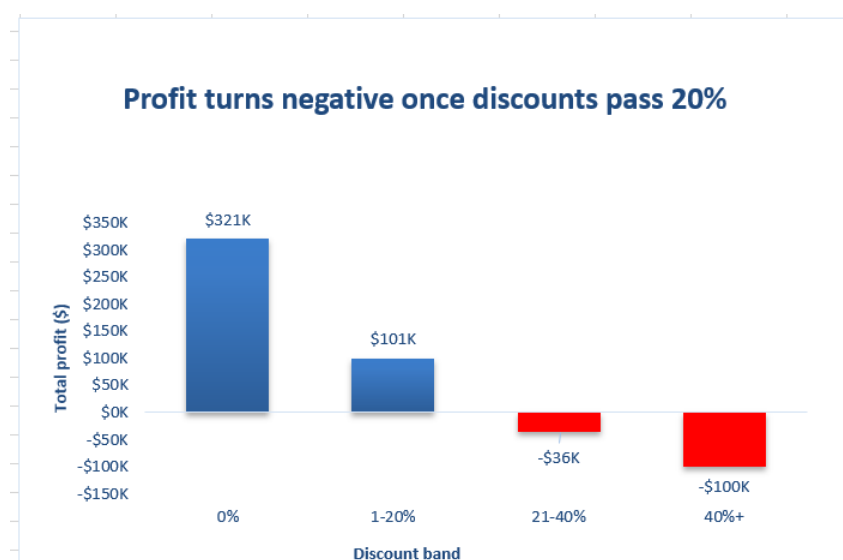
Headline: Sales are growing, but discounts above 20% are wiping out profit.

Order lines discounted 21% or more lost \$135K in total, while undiscounted lines earned \$321K.

Business questions

1. Which sub-categories and regions make money, and which lose money?
2. Does the discount level explain the losses? (headline question)
3. How did sales change from 2014 to 2017, and which months are strongest?
4. Which regional manager's region earns the most profit, and at what margin?

Chart 1: Profit by discount band



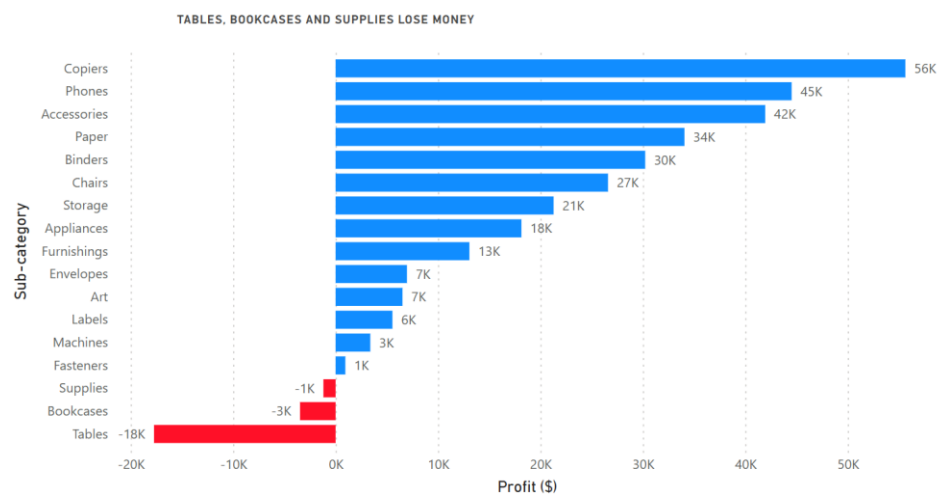
Profit turns negative once discounts pass 20%: the 1,393 order lines discounted above 20% lost \$135K, while all other lines earned \$422K.

So what: Cap discounts at 20% and require approval for anything higher, starting with Tables, the biggest loss-maker.

This is the clearest place to recover profit without hurting sales growth.

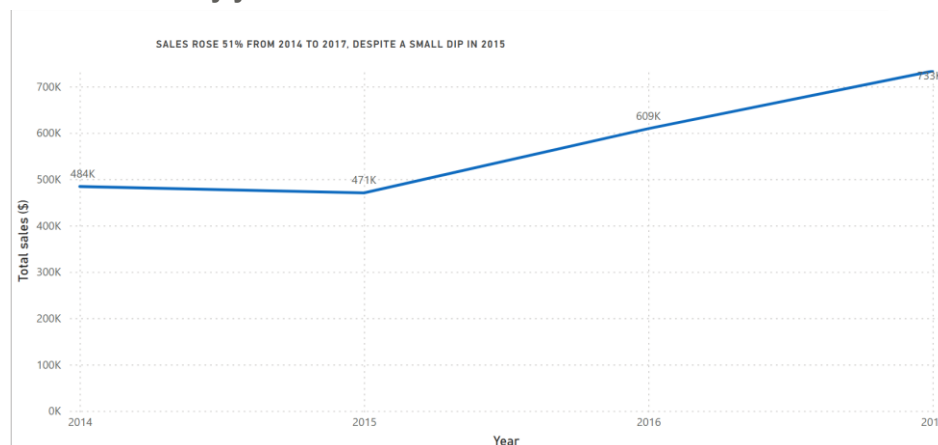
Supporting evidence

Chart 2: Profit by sub-category



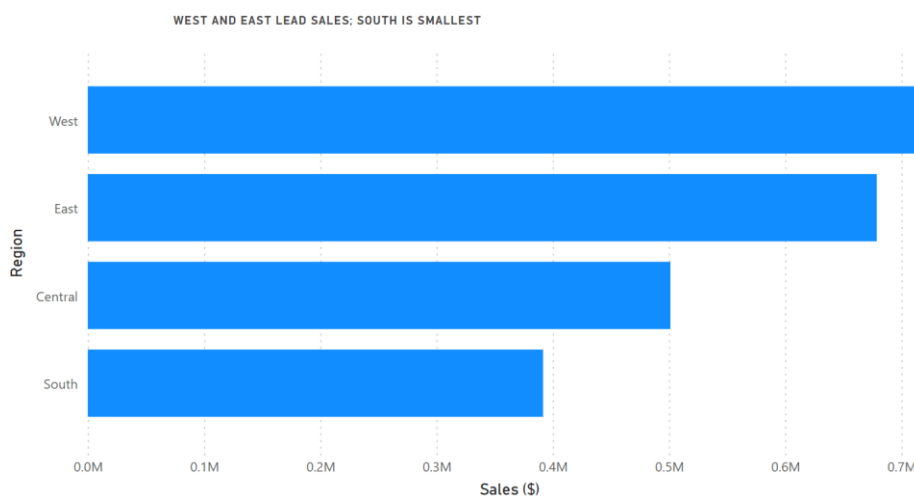
Three sub-categories lose money, led by Tables at -\$17.7K, while Copiers earn the most at \$56K.

Chart 3: Sales by year



Sales rose 51%, from \$484K in 2014 to \$733K in 2017, with a small dip in 2015.

Chart 4: Sales by region



West (\$725K) and East (\$679K) lead sales and South (\$392K) is smallest. Central keeps about 8 cents of profit per sales dollar, versus 12 to 15 cents in the other regions.

Caveats

- This is a sample dataset covering 2014 to 2017, not live company data.
- 5,201 rows had dates corrected (month and day were swapped in some order dates and ship dates). After the fix, every shipping time is 0 to 7 days..
- Counts by discount band are order lines (9,994 rows), not unique orders (5,009).
- The data shows heavy discounts and losses occurring together. It does not prove discounts alone cause the losses; product mix may also play a part.

Honesty check

Total profit reported: \$286,397. **Raw data check:** SUM(Profit) on the cleaned dataset is 286,397.02, and the four discount bands in the PivotTable also add up to 286,397.02. The numbers match.

Data Profiling (Phase 1)

Dataset: Sample Superstore, 2014-2017. 9,994 order lines, 5,009 unique orders, 26 columns after cleaning. No missing values; 1 possible duplicate pair (see note below the table).

Column	Type	Missing	Distinct	Issues found
Row_ID	Integer	0	9,994	Unique, none
Order_ID	Text	0	5,009	Repeats by design (several lines per order)
Ship_Mode	Text	0	4	None
Customer_ID / Customer_Name	Text	0	793	Odd non-breaking spaces in some names
Segment	Text	0	3	None
Country / City / State	Text	0	1 / 531 / 49	None
Postal_Code	Text	0	631	449 codes lost their leading zero
Region	Text	0	4	None
Product_ID / Product_Name	Text	0	1,862 / 1,850	Odd spaces in names
Category / Sub_Category	Text	0	3 / 17	None
Sales	Decimal	0	5,825	Range 0.44 to 22,638
Quantity	Integer	0	14	Range 1 to 14
Discount	Decimal	0	12	Range 0% to 80%
Profit	Decimal	0	7,287	1,871 lines are negative (loss-making)
Order_Date / Ship_Date	Date	0	1,237 / 1,334	Mixed formats, month and day swapped

Table 1: Profiling of the raw data.

Duplicate check: 1 possible duplicate pair (order US-2014-150119, different Row_IDs). It was kept because it could be the same item ordered twice, and its effect on the totals is negligible.

Cleaning log (Phase 2)

Problem	What I did	Rows affected
Mixed date formats: some dates were read day-first, so month and day were swapped	Swapped them back on real-date cells and read text dates as month/day/year	5,201 rows (4,042 order-date and 3,898 ship-date cells)
Shipping days wrong (up to 214 days) because of the swap	Recalculated after the fix; every shipping time is now 0 to 7 days	All 9,994 rows
Postal codes lost the leading zero (e.g. 6824)	Stored as text and padded to 5 digits	449 rows
Non-breaking spaces in customer and product names	Replaced with normal spaces	225 rows
Stray blank rows and a check value at the bottom of the file	Removed	2 rows
Column names with spaces and hyphens	Renamed with underscores (e.g. Sub-Category to Sub_Category)	9 columns
Analysis columns missing	Added Shipping_Days, Order_Year, Order_Month, Profit_Margin, Discount_Band	All rows

Table 2: Cleaning log (problem, action, rows affected).

Verification after cleaning and loading into MySQL: 9,994 rows, total profit \$286,397.02, shipping days 0 to 7, order dates from 2014-01-03 to 2017-12-30, and orders per year of 1,993 / 2,102 / 2,587 / 3,312.

Before and after: 5 date rows

Row ID	Raw order	Raw ship	Fixed order	Fixed ship	Shipping days
1	11-08-2016	11-11-2016	2016-11-08	2016-11-11	3
2	11-08-2016	11-11-2016	2016-11-08	2016-11-11	3
19	8/27/2014	09-01-2014	2014-08-27	2014-09-01	5
3	06-12-2016	6/16/2016	2016-06-12	2016-06-16	4
13	4/15/2017	4/20/2017	2017-04-15	2017-04-20	5

Table 3: Before and after for 5 date rows. Raw dates in dd-mm-yyyy form were read in the wrong order; m/d/yyyy ones were text.

SQL JOIN (Phase 3)

Question: Which regional manager's region earns the most profit, and at what margin?

A small reference table, regional_managers (Region, Manager), was created to demonstrate a JOIN across two tables. The manager names are placeholders.

```
SELECT m.Manager,
       o.Region,
       ROUND(SUM(o.Sales), 2) AS total_sales,
       ROUND(SUM(o.Profit), 2) AS total_profit,
       ROUND(SUM(o.Profit) / SUM(o.Sales) * 100, 1) AS margin_pct
FROM orders o
JOIN regional_managers m ON o.Region = m.Region
GROUP BY m.Manager, o.Region
ORDER BY total_profit DESC;
```

Manager	Region	Total sales (\$)	Total profit (\$)	Margin (%)
Manager D	West	725,457.82	108,418.45	14.9
Manager B	East	678,781.24	91,522.78	13.5
Manager C	South	391,721.91	46,749.43	11.9
Manager A	Central	501,239.89	39,706.36	7.9
Total		2,297,200.86	286,397.02	

Table 4: JOIN result. The profit total matches the honesty check (\$286,397.02), so the join lost and duplicated no rows.

The screenshot displays the MySQL Workbench interface. The top pane shows a SQL query that joins the 'orders' table with the 'regional_managers' table. The query calculates total sales, total profit, and margin percentage for each manager, ordered by total profit in descending order. The bottom pane shows the 'Result Grid' with the query's output. The result grid has columns for Manager, Region, total_sales, total_profit, and margin_pct. The data rows show Manager D (West) with the highest profit, followed by Manager B (East), Manager C (South), and Manager A (Central) with the lowest profit. The total profit across all managers is \$286,397.02.

```
1 • SELECT m.Manager,
2       o.Region,
3       ROUND(SUM(o.Sales), 2) AS total_sales,
4       ROUND(SUM(o.Profit), 2) AS total_profit,
5       ROUND(SUM(o.Profit) / SUM(o.Sales) * 100, 1) AS margin_pct
6 FROM orders o
7 JOIN regional_managers m ON o.Region = m.Region
8 GROUP BY m.Manager, o.Region
9 ORDER BY total_profit DESC;
```

Manager	Region	total_sales	total_profit	margin_pct
Manager D	West	725457.82	108418.45	14.9
Manager B	East	678781.24	91522.78	13.5
Manager C	South	391721.91	46749.43	11.9
Manager A	Central	501239.89	39706.36	7.9

Figure: JOIN query and result in MySQL Workbench. The four profit values add up to \$286,397.02.

Reading the result: Central has the second-highest sales but the lowest margin (7.9%), while West earns 14.9%. Central also has the heaviest discounting: an average discount of 24.0% and 27.7% of its order lines discounted above 20%, against 3.7% in West. This fits the discount finding, but it shows correlation, not proof of cause.

When SQL was easier than Excel: one query linked two tables and totalled 9,994 rows, where Excel would need a lookup column plus a pivot. The query can also be re-run unchanged if the data is refreshed.

Seasonality

Sales are strongly seasonal. November (\$352K), December (\$325K) and September (\$308K) are the strongest months across the four years, and Q4 (October to December) brings in about 38% of all sales. January (\$95K) and February (\$60K) are the weakest. Four years of sample data show the pattern but not its cause.

Explore next

Do Bookcases and Supplies lose money for the same reason as Tables (heavy discounting), and what discount level is the break-even point for each sub-category?

The story

Sales grew 51% between 2014 and 2017, so the business looks healthy at first glance. But growth is not the same as profit. Across the dataset, order lines with no discount earned \$321K and lines with a 1 to 20% discount earned \$101K, while lines discounted more than 20% lost \$135K. The same pattern shows up by product: Tables, the biggest loss-maker, lost \$17.7K. Regional differences are smaller than the discount effect, so the recommendation is to cap discounts rather than to change regional strategy.

Why each chart

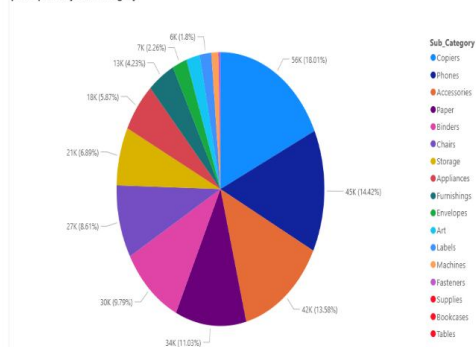
Insight	Question it answers	Number behind it	Chart and why
1. Discounts vs profit	Does heavy discounting hurt profit?	Lines discounted above 20% lost \$135K; undiscounted lines earned \$321K	Column chart of profit by discount band: compares groups and shows negative values clearly.
2. Sub-category profit	Which products lose money?	Tables -\$17.7K, Bookcases -\$3.5K, Supplies -\$1.2K	Sorted bar chart: 17 long labels, negatives visible, ranking obvious.
3. Yearly sales	Is the business growing?	\$484K (2014) to \$733K (2017), +51%	Line chart: a trend over time is best shown as a line.
4. Regions	Which regions lead?	West \$725K, East \$679K, Central \$501K, South \$392K	Bar chart: a simple comparison of four categories.

Write-up: chart comparisons

Wrong chart vs right chart

Wrong: pie of profit by sub-category

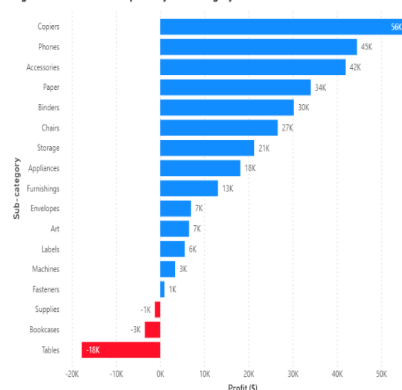
pie of profit by sub-category



The pie can't show negative profit, so it silently drops the three loss-making sub-categories, and 14 slices are hard to rank.

Right: sorted bar of profit by sub-category

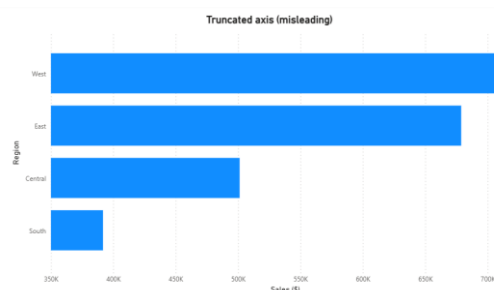
Right chart: sorted bar of profit by sub-category



The sorted bar ranks all 17 sub-categories and shows the three losses: Tables (-\$18K), Bookcases (-\$3K) and Supplies (-\$1K).

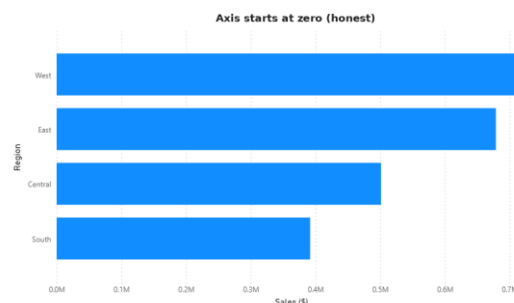
Bad vs honest chart

Bad: truncated axis (starts at \$350K)



Truncated axis: South looks about nine times smaller than West.

Honest: axis starts at zero



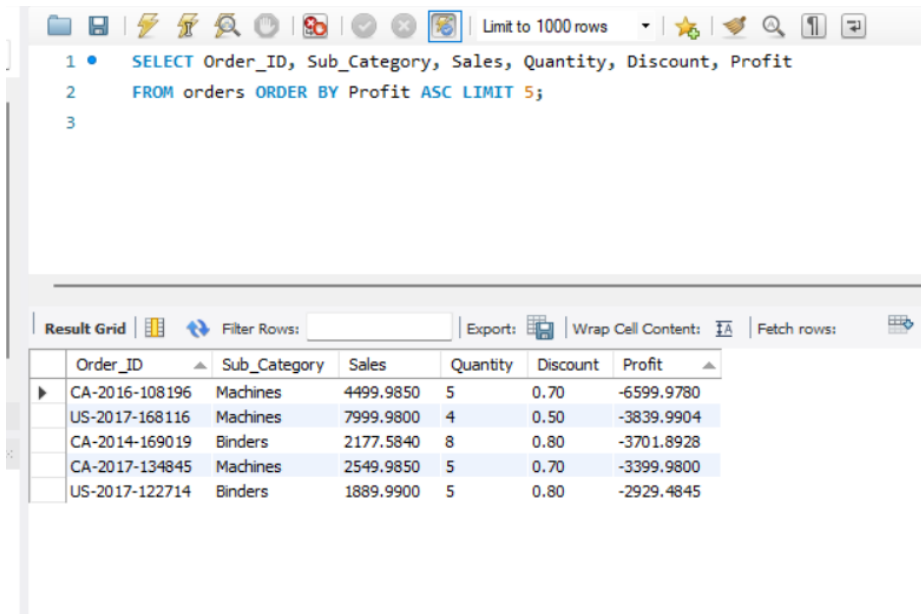
Zero-based axis: West (\$725K) is about 1.85 times South (\$392K).

Two rules for an honest chart

- **Start bar-chart axes at zero.** Bar length is read as size, so cutting the axis exaggerates differences.
- **Label everything so the chart stands alone.** Give it a title that states the finding, axis titles, and units.

Outlier check

The 5 biggest-loss order lines (losses from \$2,929 to \$6,600) are all Machines or Binders sold at 50-80% discount. Quantities and sales values are normal, so these are real business losses caused by deep discounting, not data errors. Together they lost \$20,471, about 7% of total profit (\$286,397). This supports the recommendation to cap discounts at 20%.



The screenshot shows a SQL query editor with the following query:

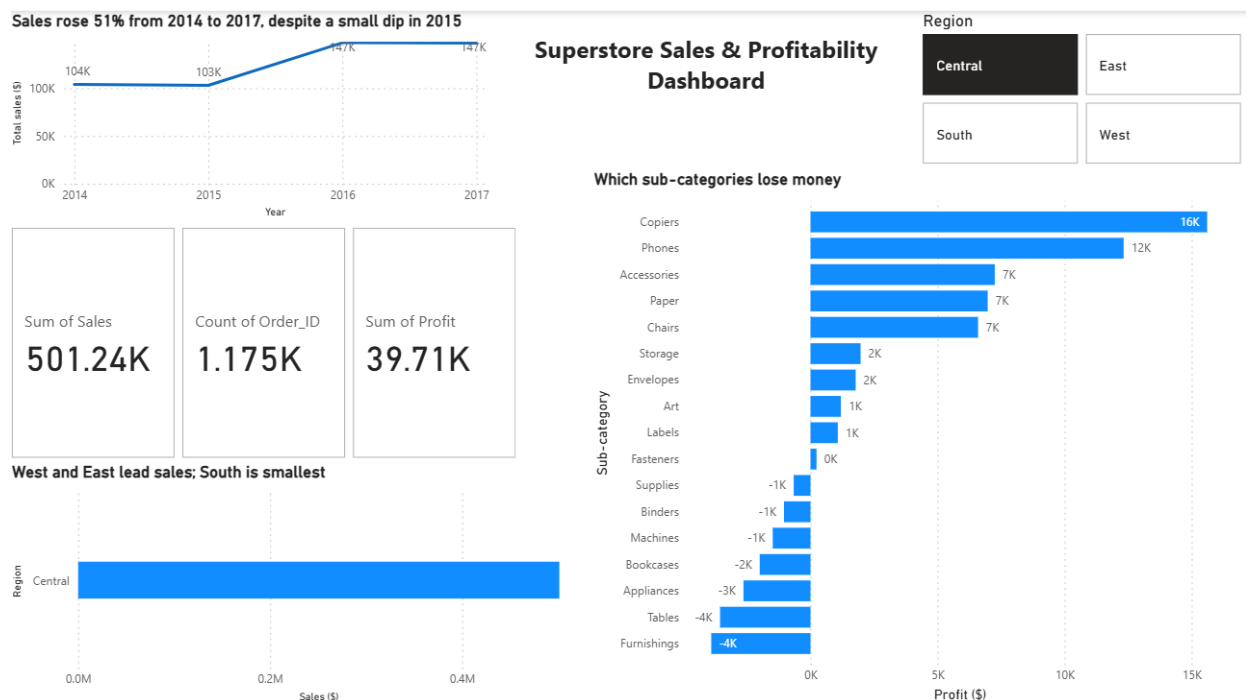
```
1 • SELECT Order_ID, Sub_Category, Sales, Quantity, Discount, Profit
2 FROM orders ORDER BY Profit ASC LIMIT 5;
3
```

Below the query is a 'Result Grid' showing the results of the query. The grid has columns for Order_ID, Sub_Category, Sales, Quantity, Discount, and Profit. The data is as follows:

Order_ID	Sub_Category	Sales	Quantity	Discount	Profit
CA-2016-108196	Machines	4499.9850	5	0.70	-6599.9780
US-2017-168116	Machines	7999.9800	4	0.50	-3839.9904
CA-2014-169019	Binders	2177.5840	8	0.80	-3701.8928
CA-2017-134845	Machines	2549.9850	5	0.70	-3399.9800
US-2017-122714	Binders	1889.9900	5	0.80	-2929.4845

The 5 biggest-loss order lines, found with ORDER BY Profit ASC.

Dashboard interactivity (slicer)



Dashboard filtered to Region = Central, showing the slicer is interactive. Bars to the left of zero are loss-making sub-categories. Chart titles describe the full dataset; the numbers update with the filter.

Written answers

1. Choosing the right chart type

Start with the message, then match the chart to it:

- **Comparison:** bar or column chart. Salesforce's design guidelines call the bar chart the workhorse of analytics, because our eyes quickly see differences in length.
- **Trend:** line chart. Qlik's guidance lists line charts for comparing trends over time.
- **Composition:** pie, donut or stacked bar, used only for a few parts. Qlik describes pie and donut charts as suited to the static composition of a value.
- **Relationship:** scatter plot, which shows how two or three measures relate.

The Salesforce guidelines also warn that the wrong chart type risks misrepresenting the data and confusing or misleading the audience.

Sources: Salesforce Lightning Design System, "Charts" guidelines (lightningdesignsystem.com/guidelines/charts); Qlik Support, "Best practices for choosing visualization types" (support.qlik.com/articles/000057453).

2. Two ways charts can mislead

Truncated axis. If a bar chart's axis starts at \$350K instead of \$0, a small difference looks huge. Fix: start bar axes at zero and label them.

Misleading pie chart. Pies with many slices, similar sizes or negative values are hard to read, and a pie cannot show a loss. In this project it dropped Tables, Bookcases and Supplies entirely. Fix: use a sorted bar chart, and keep pies for a few parts that add up to 100%.

3. What makes data storytelling effective

Showing charts just displays data and leaves the reader to work out what matters. Effective storytelling has one clear message the reader can repeat, a logical order (headline first, then evidence, then what to do), insight written in words (titles like "Profit turns negative once discounts pass 20%" instead of "Sum of Profit by Band"), focus with clutter removed, and a clear so-what for a specific audience.

4. Why lead with the main insight (manager note)

Managers are short on time and decide based on what matters, not how you got there. Leading with the main insight lets them understand the point in seconds and judge whether it needs action. It also makes the rest of the report easier to follow, since every chart then reads as evidence for a conclusion they already know. Starting with methodology forces a busy reader to wade through detail before learning why it matters, and they may stop reading before the key finding. The methodology still belongs in the report, but as a supporting appendix for anyone who wants to check it. In short: answer first, evidence next, method last.